

FinPilot

AI-Powered Investment Analysis Platform Business Plan 2026

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- Target Programs**
- FFG Basisprogramm 2026
 - aws Gründungsfonds II
 - EIC Accelerator 2026
 - NSF SBIR Phase I
 - riz-up GENIUS 2026

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1. Company Overview

FinPilot is an AI-powered investment intelligence platform developed in Gänserndorf, Lower Austria. The platform combines Deep Reinforcement Learning (DRL) agents with a multi-provider Large Language Model (LLM) router to deliver automated stock analysis across 1,148 NASDAQ symbols spanning 11 market sectors.

1.1 Mission Statement

To democratise algorithmic investment analysis by making institutional-grade AI technology accessible to individual investors, small funds, and fintech developers — at a fraction of the cost of legacy solutions.

1.2 Company Details

Field	Details
Company Name	FinPilot
Location	Gänserndorf, Niederösterreich, Austria
Founded	2024 (Sprint 1)
Current Stage	Sprint 22 — Advanced Prototype / Pre-Revenue
Platform Version	v1.7.0
Contact	mericbsk@gmail.com
Legal Form	To be established (GmbH / Einzelunternehmen)
Target Markets	Austria, European Union, United States

1.3 Development History

FinPilot has been developed through a rigorous sprint-based methodology, with 22 completed development cycles. Each sprint delivered tangible, testable functionality:

- Sprints 1–8: Core scanner, data pipeline, Streamlit dashboard, Telegram notifications.
- Sprints 9–14: DRL agent training (PPO/SAC/TD3/A2C), walk-forward validation, model registry.
- Sprints 15–18: Reward shaping, curriculum learning, PilotShield risk control, Optuna HPO.
- Sprints 19–22: LLM Router (Groq → Claude → Gemini), FastAPI backend, JWT auth, PostgreSQL migration, CI/CD hardening.

2. Problem & Solution

2.1 The Problem

Retail investors face a fundamental information asymmetry compared to institutional players. While hedge funds and investment banks deploy sophisticated algorithmic tools, individual investors are left with basic charting applications that provide indicators but not intelligent decision support.

Key Pain Points

- **Cost barrier:** Professional systems (Bloomberg Terminal €25K/year, FactSet €30K/year) are inaccessible to 99% of retail investors.
- **Complexity barrier:** Existing algorithmic tools (QuantConnect, Zipline) require advanced programming skills.
- **Information overload:** Over 5,000 tradeable symbols on major exchanges; manual analysis is impossible.
- **No adaptive learning:** Static rule-based indicators do not learn from market regime changes.
- **Fragmented intelligence:** Technical analysis, fundamental data, sentiment, and on-chain signals exist in separate silos.

2.2 Our Solution

FinPilot integrates five previously fragmented intelligence layers into a single, automated platform:

Intelligence Layer	FinPilot Implementation	Replaces
Technical Analysis	Scanner: 1,148 symbols, 11 sectors, EMA/RSI/MACD/BB/ATR	TradingView (€15/month)
Adaptive Learning	19 DRL agents (PPO/SAC/TD3/A2C) with walk-forward retraining	Bloomberg (€25K/year)
LLM Synthesis	Groq → Claude → Gemini router, automatic failover, 99.9% uptime	Manual analyst reports
Backtesting	Historical backtesting engine, Optuna HPO, 30 trials/agent	QuantConnect (€200/month)
Risk Management	PilotShield: position limits, leverage controls, confidence thresholds	Custom risk systems

2.3 Unique Value Proposition

FinPilot is the only platform that combines DRL-based adaptive signal generation, multi-LLM synthesis, and an enterprise-grade production stack at a consumer-accessible price point. Competitive analysis shows no direct equivalent in the €0–€100/month retail segment.

3. Product

3.1 Platform Architecture

FinPilot is built as a cloud-native, microservices-ready platform with clear separation of concerns across frontend, API, ML engine, data pipeline, and infrastructure layers.

Layer	Technology	Status
Dashboard (Primary)	Streamlit v1.7.0 — 5-tab analysis interface	☐ Production
Frontend (Next-gen)	Next.js + TypeScript — modern responsive UI	⚠ In Progress
Backend API	FastAPI — 11 routers, JWT auth, SlowAPI rate limiting	☐ Production
DRL Engine	Stable-Baselines3 PPO/SAC/TD3/A2C + Gymnasium env	☐ Active
LLM Router	Groq → Claude → Gemini (automatic failover)	☐ Production
Hyperparameter Opt.	Optuna — 30 trials/agent, SQLite/PostgreSQL	☐ Active
Database	SQLite (dev) / PostgreSQL (production)	⚠ Migration Pending
Broker Integration	Alpaca API — paper & live trading	⚠ Paper mode
Monitoring	Prometheus + Grafana + Sentry error tracking	☐ Active
Infrastructure	Docker Compose + GitHub Actions CI/CD (7 jobs)	☐ Production
Security	JWT HS256, bcrypt, detect-secrets, Bandit/Safety	☐ Active

3.2 DRL Agent Stack

The proprietary DRL engine is the core technological differentiator. FinPilot trains agents using a Gymnasium-compatible market simulation environment with the following design principles:

- Walk-forward training with 70/30 train/test split prevents overfitting to historical data.
- 5-phase curriculum learning progressively increases task complexity for stable convergence.
- Composite scoring: $(\text{Sharpe Ratio} \times 10) + (\text{Calmar Ratio} \times 0.3) - \text{overtrading penalty} - \text{negative return penalty}$.
- PilotShield risk layer: max absolute position = 75%, max leverage = 1.5×, confidence threshold = 65%.
- LearnableEnsembleWeights: EMA-based dynamic agent weighting combining regime prior and performance EMA.
- Autonomous improvement loop: `drl_autopilot.py` weekly cycle measures, analyses, recommends, and generates retrain commands.

3.3 Current Model Performance

Model ID	Type	Return	Max DD	Sharpe	Score
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ppo_trend_20260304	Trend	+574%	46.5%	0.057	4.27
ppo_trend_20260305	Trend	+353%	37.7%	0.053	3.33
ppo_momentum_20260306	Momentum	+300%	47.3%	0.063	2.23
ppo_trend_20260304b	Trend	+270%	46.8%	0.043	2.16
ppo_momentum_20260303	Momentum	+211%	41.1%	0.055	1.80

Active optimisation in progress: dd_weight parameter increase (0.3 → 0.8) targeting MaxDD <20% and Sharpe >0.5 within 2 iterations.

3.4 Product Roadmap

Quarter	Milestone	Priority
Q2 2026	Cloud deployment (AWS/Azure), PostgreSQL migration, paid beta launch	Critical
Q2 2026	WebSocket real-time data feeds activation	High
Q3 2026	Next.js frontend completion and Streamlit deprecation	High
Q3 2026	Mobile app (React Native) beta release	Medium
Q4 2026	API marketplace and B2B onboarding portal	High
Q1 2027	DRL model targets: MaxDD <20%, Sharpe >0.5	Critical
Q2 2027	First white-label deployment for institutional client	High
Q3 2027	Series A readiness: 1,000+ paying subscribers, 5 B2B contracts	Strategic

4. Market Analysis

4.1 Total Addressable Market (TAM)

Global Algo Trading Market (2028)	\$31.4 Billion	CAGR	11.6%
EU Retail Investors	58 Million+	Using Algo Tools	<2%
AI in Finance Market (2030)	\$22.6 Billion	CAGR	23.4%
Austrian Retail Investors	2.5 Million	Digital First	45%

4.2 Serviceable Addressable Market (SAM)

FinPilot’s SAM is defined as EU-based individual investors and small funds actively seeking algorithmic support tools, plus fintech developers requiring AI stock analysis APIs.

- EU active retail traders: ~4.2 million (7.2% of 58M retail investors).
- European fintech startups requiring financial AI APIs: ~12,000 companies.
- Austrian-DACH market as primary beachhead: ~180,000 active algo-interested investors.

4.3 Serviceable Obtainable Market (SOM)

Conservative 3-year market capture estimate based on comparable SaaS tools in the financial intelligence space (e.g., Koyfin, Finviz Premium):

Year 1 Subscribers	450	Year 1 Revenue	€85,000
Year 2 Subscribers	2,200	Year 2 Revenue	€340,000
Year 3 Subscribers	6,000	Year 3 Revenue	€1,200,000

4.4 Competitive Landscape

Competitor	Strengths	Weaknesses	FinPilot Advantage
Bloomberg Terminal	Data depth, institutional trust	€25K/year, no DRL, no LLM	100× lower cost, adaptive AI
TradingView	Large community, charting	No AI decision support, no DRL	DRL + LLM intelligence layer
QuantConnect	Algorithmic backtesting	Requires programming skills	No-code, automated signals
Koyfin	Modern UI, data breadth	No ML predictions	Predictive DRL signals
Finviz Premium	Fast screener	Rule-based only, no learning	Adaptive learning agents

4.5 Regulatory Environment

FinPilot operates within the EU’s MiFID II regulatory framework. Key positioning decisions:

- Platform is positioned as an 'investment research and analysis tool', NOT an 'investment adviser'.
- All signals are accompanied by clear disclaimers: "For informational purposes only. Not financial advice."
- User authentication and access control (JWT) ensures trackable usage for compliance purposes.
- FMA (Austrian Financial Market Authority) advisory consultation planned for Q2 2026.
- Data residency: EU-based cloud infrastructure preferred for GDPR compliance.

5. Business Model

5.1 Revenue Streams

Stream	Model	Price	Year 1 Target	Year 3 Target
SaaS — Free	Freemium acquisition	€0	2,000 users	20,000 users
SaaS — Pro	Monthly subscription	€29/mo	300 users	3,500 users
SaaS — Premium	Monthly subscription	€99/mo	80 users	1,200 users
B2B API	Usage-based	€49/10K calls	5 clients	40 clients
White-label	Project-based	€5K–20K/project	1 project	8 projects
Education/Community	Monthly membership	€15–25/mo	200 members	2,000 members

5.2 Cost Structure

Cost Category	Year 1	Year 2	Year 3
Cloud Infrastructure	€18,000	€36,000	€72,000
Third-party APIs (LLM, Data)	€12,000	€24,000	€48,000
Personnel (1 FTE + contractors)	€45,000	€90,000	€180,000
Marketing & Sales	€8,000	€25,000	€60,000
Legal & Compliance	€5,000	€10,000	€15,000
R&D (DRL / product)	€0	€20,000	€40,000
Total OPEX	€88,000	€205,000	€415,000

5.3 Unit Economics

CAC (Customer Acquisition Cost)	€18–35	LTV (Lifetime Value, Pro)	€420
LTV/CAC Ratio	12–23×	Gross Margin (SaaS)	~78%
Payback Period	~2 months	Churn Target	<5%/month
Break-even (Monthly)	Month 18	Break-even Users	~1,400

6. Go-to-Market Strategy

6.1 Phase 1: Beachhead (Q2–Q3 2026)

Focus on DACH (Germany, Austria, Switzerland) retail investors and developer community. Key channels:

- Content marketing: “How we built a DRL trading platform in 22 sprints” — developer blog and YouTube series.
- Open-source strategy: Core scanner module open-sourced on GitHub to build developer trust and community.
- riz-up network: Leverage Gänserndorf consulting ecosystem for early B2B introductions.
- Product Hunt launch: Targeting “Product of the Day” in the Finance Tools category.
- Communities: QuantConnect forum, r/algotrading, LinkedIn algorithmic trading groups.

6.2 Phase 2: European Expansion (Q4 2026 – Q2 2027)

- EIC Accelerator funding enables pan-European marketing push.
- Localisation: German, French, Spanish platform versions.
- Partnership with 2–3 European fintech accelerators (e.g., FinTech Austria, Berlin Fintech Week).
- Press coverage: TechCrunch Europe, Der Standard (Austria), Handelsblatt (Germany).

6.3 Phase 3: US Entry (2027)

- NSF SBIR funding establishes US credibility and non-dilutive capital.
- US entity formation (Delaware C-Corp) for investor compatibility.
- API-first US strategy: Target algorithmic trading developers on RapidAPI marketplace.
- Potential US accelerators: Y Combinator, Techstars NYC, FinTech Innovation Lab.

7. Team

7.1 Founder

The founding team brings together deep technical expertise across three critical domains required for FinPilot’s success: AI/ML engineering, financial markets, and full-stack software development.

Competency	Evidence
AI/ML Engineering	19 DRL models trained, Optuna HPO, walk-forward validation, curriculum learning
Financial Markets	Deep understanding of Sharpe, Calmar, drawdown, market regime detection
Software Engineering	22-sprint production system: FastAPI, Docker, CI/CD, JWT, Prometheus
Product Development	7 core modules, 16 components, 466 tests, enterprise-grade architecture
Research Discipline	Systematic experiment logging, research state management, regression detection

7.2 Hiring Plan (Post-Funding)

- Month 3: Full-stack engineer (Next.js / React Native) — frontend completion and mobile app.
- Month 6: DevOps / Cloud engineer — AWS deployment, auto-scaling, PostgreSQL migration.
- Month 9: Sales / Business Development — B2B pipeline and institutional partnerships.
- Month 12: Compliance officer (part-time) — MiFID II, FMA advisory, GDPR.

7.3 Advisory Board (Planned)

- Quantitative Finance Expert — DRL model validation and financial methodology review.
- Regulatory Advisor (Austria/EU) — MiFID II compliance, FMA liaisons.
- Fintech Entrepreneur (DACH) — go-to-market strategy and investor introductions.

8. Financial Plan

8.1 Revenue Projections

Revenue Stream	Year 1 (2026)	Year 2 (2027)	Year 3 (2028)
SaaS Pro (300/2,200/6K users)	€87,600	€259,600	€612,000
SaaS Premium (80/400/1.2K users)	€47,520	€237,600	€713,000
B2B API (5/15/40 clients)	€14,700	€44,100	€117,600
White-label (1/3/8 projects)	€10,000	€45,000	€120,000
Education / Community	€18,000	€60,000	€150,000
TOTAL REVENUE	€177,820	€646,300	€1,712,600
TOTAL COSTS	€88,000	€205,000	€415,000
GROSS PROFIT	€89,820	€441,300	€1,297,600

8.2 Funding Requirements

FinPilot is seeking €500,000 in funding through a combination of grants, government-backed investments, and strategic angel investment. The funding will be deployed over 18 months to reach cash-flow positive operations.

Category	Amount	Purpose	Timeline
Cloud Infrastructure	€80,000	AWS/Azure deployment, scaling	Month 1–3
AI/ML R&D	€150,000	DRL optimisation, real-time data integration	Month 1-2
Product Development	€120,000	Next.js, mobile, API marketplace	Month 3-9
Legal & Regulatory	€50,000	MiFID II, FMA, IP protection	Month 1–6
Marketing & Sales	€70,000	Launch campaigns, B2B outreach	Month 4–12
Operations	€30,000	Team, administration, tools	Month 1–12
TOTAL	€500,000	—	18 months

8.3 Grant Funding Strategy

Program	Region	Amount	Probability	Timeline
riz-up GENIUS 2026	Austria	€3,000–10,000	High	September 2026
FFG Basisprogramm	Austria	€50K–200K	Medium–High	Q3 2026
aws Preseed	Austria	€25K–50K	Medium	Q2–Q3 2026
EIC Accelerator	EU	€250K–2.5M	Medium	Q3–Q4 2026
NSF SBIR Phase I	USA	€280K (≈\$305K)	Medium	Q4 2026

9. Risk Analysis & Mitigation

Risk	Likelihood	Impact	Mitigation
DRL model performance below expectations	Medium	High	Automated optimisation loop (drl_autopilot.py); weekly retrain cycle; ensemble diversity
Regulatory changes (MiFID II / FMA)	Low	High	Platform framed as 'research tool'; FMA advisory from Day 1; legal counsel budgeted
Key dependency failure (yfinance / LLM APIs)	Medium	Medium	Multi-provider LLM router; yfinance → Polygon.io fallback; redundant data sources
Competition from well-funded players	Medium	Medium	Speed advantage; DRL differentiation; community moat; lower price point
Cloud cost overruns	Low	Medium	Rate limiting; caching layer (Redis); auto-scaling with hard spend caps
Key person risk (solo founder)	High	High	Hiring plan from Month 3; documentation discipline; structured codebase

10. Appendix

10.1 Technology Stack Summary

Component	Technology	Version / Notes
Primary Frontend	Streamlit	v1.7.0, 5-tab dashboard
Secondary Frontend	Next.js + TypeScript	In migration
Backend API	FastAPI + Uvicorn	11 routers, JWT, SlowAPI
DRL Framework	Stable-Baselines3	PPO, SAC, TD3, A2C
RL Environment	Gymnasium	Custom MarketEnv
HPO Engine	Optuna	30 trials/agent
LLM Provider 1	Groq	Primary (speed)
LLM Provider 2	Anthropic Claude	Secondary (quality)
LLM Provider 3	Google Gemini	Tertiary (fallback)
Database	SQLite / PostgreSQL	Migration in progress
Broker	Alpaca	Paper + Live API
Monitoring	Prometheus + Grafana + Sentry	Active stack
CI/CD	GitHub Actions	7 jobs: test, lint, security, docker, scanner, drl, frontend
Containerisation	Docker Compose	Multi-profile, multi-service
Security	JWT HS256 + bcrypt + detect-secrets + Bandit	Production-grade

10.2 Key Metrics Summary (March 2026)

Total Test Functions	466	Test Files	23
Registered DRL Models	19	Active CI/CD Jobs	7
Scanned Symbols	1,148	Market Sectors	11
API Endpoints	11	Third-party Integrations	12
Sprint Cycles Completed	22	Platform Version	v1.7.0
Top Model Return (backtest)	+574%	Top Model Composite Score	4.27